ChE422-Biochemical Engineering–Spring 2019
Student Learning Outcome 1
Rubric for Assignment 2
Protein engineering

SLO 1 Performance Indicator (PI)	**(1)** Unsatisfactory	**(2)** Developing	**(3)** Satisfactory
6. Solutions creativity alternatives	Demonstrates creative synthesis of solution and creates new alternatives by combining knowledge and information (< 6 pts)	Demonstrates solution with integration of diverse concepts or derivation of useful relationships involving ideas covered in course concepts; however, no alternative solutions are generated (6-7 pts)	Demonstrates creative synthesis of solution and creates new alternatives by combining knowledge and information (8-10 pts)

Averages:

Performance Indicators: (6) 3   Overall Average: 3

Average scores were fairly uniform, all students were strong in formulating the engineering system and model equations, and a majority 8/8 were satisfactory or higher in using appropriate resources needed to solve problems.

Assignment 2 Due on Feb 25, 2020

One researcher found one organism has a very interesting property. After many steps' purification, he was able to get a small amount of protein. Then he used Mass spectrometry (MS) to find out the sequence and the molecular weight of the protein. The protein sequence is provided in the form of single-letter amino acid code.

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  1 medaknikkg papfypledg tageqlhkam kryalvpgti aftdahievd ityaeyfems
 61 vrlaeamkry glntnhrivv csenslqffm pvlgalfigv avapandiyn erellnsmgi
121 sqptvvfvsk kglqkilnvq kklpiiqkii imdsktdyqg fqsmytfvts hlppgfneyd
181 fvpesfdrdk tialimnssg stglpkgval phrtacvrfs hardpifgnq iipdtailsv
241 vpfhhgfgmf ttlgylicgf rvvlmyrfee elflrslqdy kiqsallvpt lfsffakstl
301 idkydlsnlh eiasggapls kevgeavakr fhlpgirqgy gltettsail itpegddkpg
361 avgkvvpffe akvvdldtgk tlgvnqrgel cvrgpmimsg yvnnpeatna lidkdgwlhs
421 gdiaywdede hffivdrlks likykgyqva paelesillq hpnifdagva glpdddagel
481 paavvvlehg ktmtekeivd yvasqvttak klrggvvfvd evpkgltgkl darkireili
541 kakkggkiav
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He needs to make more protein to further characterize it. Unfortunately, he does not have enough organisms to provide him enough protein for the characterization.
You need to help him by doing the following things:

(a) Propose a method to make more protein (as detailed as possible)

(b) Tell him the molecular weight of this protein

(c) Convert the protein sequence to corresponding DNA sequence (Please specific which codon you are using)

(d) What is the putative function of this protein? (using protein blast tools in one of the provided bioinformatics databases)

(e) Give at least three applications of this protein